

Metrocoalescence revision notes

Material corrections and clarifications for the second edition

Book reviewed: Metrocoalescence, second edition (2026), by Sky Tallman.

Five revisions would strengthen the accuracy and reproducibility of the book's argument. The priorities are the responsive-zoning equations and the distinction between project size and the pace of neighborhood change. The remaining points address walking distance and the scope of the financing proposal.

References use the book's printed page numbers. Add 10 to locate the corresponding page in the supplied PDF.

1 Correct the growth function and its notation

Calculation correction | pp. 38-39

The power function on p. 38 is equivalent to $R = k + \sqrt{D}$. With $k \approx 0.02235$, this does not produce the rates stated on p. 39. At $D = 5$ it gives $R \approx 2.258$, whereas the earlier project formula defines a 5% increase as $R = 1.05$. The stated examples instead require multiplication by the square root of density, with a separate distinction between the increase and the multiplier.

To reproduce the examples, use r for the fractional increase and R for the multiplier:

$$r = k \times \sqrt{D} \quad R = 1 + r \quad U = D \times T \times r$$

Here D is the density index in dwelling-unit equivalents per acre, T is the typical block area in acres, and U is the unrounded allowance for additional residential units. With $k \approx 0.02235$, D values of 5, 10 and 20 give increases of approximately 5%, 7.1% and 10%. This also agrees with the earlier expression $U = D \times T \times (R - 1)$.

Suggested revision: replace the plus sign in the power function, use r and R consistently, and apply the stated rounding rule only after calculating U . This matters because the printed expressions currently give incompatible development allowances.

2 Normalize the commercial component by area

Definition correction | pp. 36-37 and p. 67, Figure 11

On p. 36, $C + S/2,000$ converts the number of commercial spaces C and their combined floor area S into an equivalent dwelling count. Page 37 adds that result to residential density, which is expressed in units per acre. A count and a density cannot be added without first dividing the count by the relevant block area.

Retaining the proposed commercial weighting, let N be the dwelling count and A the actual block area in acres. A consistent combined index is:

$$D_a = (N + C + S/2,000) / A$$

For example, 20 dwellings and two commercial spaces totaling 2,000 square feet on a two-acre block give $(20 + 2 + 1)/2 = 11.5$ dwelling-unit equivalents per acre. Adding the unnormalized commercial count to residential density would instead give $10 + 3 = 13$, changing the index used to set development rights.

Suggested revision: distinguish the commercial equivalent count from commercial density, and show the division by area explicitly. Figure 11 already uses area normalization in its map caption; align the central definition with that principle and explain any different conversion method used in the mapping examples.

3 Define what the rate of change controls

Implementation clarification | pp. 24-25, 34-39 and 52

The book repeatedly describes Responsive Zoning as regulating the “rate of change” and protecting neighborhoods from drastic change. The equations define the permitted size of an individual project relative to existing density. Two additional distinctions are needed to establish what that limit promises residents and how a city would apply it.

Project size and elapsed time

The text already explains that successive additions increase the density baseline: the ADU example on p. 25 and the sequence on p. 52 are explicit. The remaining question concerns several projects proceeding together, or in quick succession. The formulas specify neither a time period nor a combined threshold for pending projects.

For illustration, take a three-acre block at 10 units per acre, a typical block size of three acres, and a 10% project increment. Each project may add three units. Three projects assessed against that same baseline would add nine units, a 30% increase in the block’s original 30 units. Each project meets the individual threshold. The calculation alone does not determine whether this total occurs over one year or ten.

This does not establish that rapid growth would occur in practice. It establishes that a limit on project size is insufficient, by itself, to support a claim about the speed of aggregate neighborhood change.

Suggested revision: state whether the intended control is project scale or aggregate change over time. If it is project scale, use “maximum project increment” and qualify the claims about pace. If an aggregate pace limit is intended, specify the period, treatment of pending applications, baseline update event, and treatment of coordinated phases of one development. An annual cap need not be the preferred policy; the distinction needs to be explicit.

Typical block size and actual local impact

Page 37 deliberately uses a typical block size to give projects comparable allowances at the same density. Consequently, the percentage in the formula is not necessarily the percentage increase on the actual block. Before rounding, the actual fractional increase is:

$$\text{Actual block increase} = U / (D \times A) = r \times T / A$$

With $D = 10$, $r = 10\%$ and $T = 3$ acres, the allowance is three units. On an actual one-acre block that is a 30% increase; on a six-acre block it is 5%. Both receive the same allowance because their starting densities are equal.

Suggested revision: describe this as a standardized project allowance and state the tradeoff between comparable project sizes and comparable percentage impacts on actual blocks. If the purpose is instead a uniform cap on actual block impact, the calculation must use actual block area, or an explicitly defined surrounding area. The normalization itself is a deliberate policy choice; the accompanying claims should match it.

4 Correct the density and walking distance comparison

Quantitative correction | p. 109

The text states that the same population at half the density incurs double the average walking time and distance to the center. That is not the general geometric relationship. For a settlement of fixed population, with the same shape and a similarly scaled street network, halving density doubles area. Its linear dimensions, including distances to the center, grow by the square root of two:

$$\text{Distance multiplier} = \sqrt{2} \approx 1.414$$

Under those assumptions, average walking distance and time at an unchanged walking speed rise by about 41%, not 100%. Actual walking distances also depend on street connectivity and the location of destinations, so neither percentage should be presented as a universal empirical rule.

The same sentence says per-person infrastructure costs double. That multiplier also needs a specified network and cost model or a supporting source; it does not follow from density alone.

Suggested revision: preserve the argument that compactness can support walkability and infrastructure efficiency, replace the walking-distance multiplier with the conditional geometric relationship, and support or qualify the separate claim about infrastructure costs.

5 Clarify whom the financing proposal helps retain

Scope clarification | pp. 29-31

The book recognizes that reinvestment can raise prices and rents and create displacement pressure. Its response centers on construction finance for existing homeowners, municipal support for small projects, and access to capital for incremental development. These mechanisms explain how resident owners could participate in reinvestment and gain an alternative to selling.

They do not, by themselves, explain how existing renters remain in the neighborhood when rents rise or their buildings are redeveloped. The discussion of Baugruppen on p. 31 broadens the ownership model, but does not establish how existing renters with limited capital would gain access to it. The book therefore leaves part of its stated aim to “avoid patterns of gentrification” on p. 29 unresolved.

The relevant distinction is between enabling resident owners to participate in redevelopment and enabling existing renters to remain. These outcomes can diverge. The proposed financing mechanism establishes a path for the first; a path for the second still needs to be explained. This is a limit on the argument’s scope, not evidence that the proposed reforms would necessarily increase displacement.

Suggested revision: qualify the community-stability claim to reflect the people and mechanisms actually covered. If the broader claim is retained, explain the route by which existing renters benefit and the evidence needed to judge it. The financing proposal can stand on its merits while its contribution to renter retention is stated more precisely.